Pathri Vidya Praveen

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.23
Narayana Junior College,Vijayawada	Class XII (BIEAP)	2024	98.70%
Dr KKR Gowtham High School,Vijayawada	Class X (BSEAP)	2022	97.83%

SCHOLASTIC ACHIEVEMENTS

- Received **Academic Excellence Award (2025)** for securing the highest Semester-1 CGPA **(9.88)** in the CSE'28 batch.
- Achieved **Specialist on Codeforces** (Codeforces Profile).
- Secured **A+ grade** in Information Theory - Coding and Inference, Calculus-I, Calculus-II.
- Received **Certificate of Merit for NSEC Olympiad 2023** for placing in National Top 1% of Participants.

INTERNSHIP EXPERIENCE

Cryptography Research Intern at CCS, IITH:

- Conducted theoretical research on **Zero-Correlation Linear Cryptanalysis of unbalanced Feistel networks**, developing mathematical frameworks for cryptanalytic attacks.

RESEARCH EXPERIENCE

Computational Optimal Transport for Flow and Diffusion Models guided by Prof. Sakethanath Jagarlapudi, Dept. of CSE:

- Working on Optimal Transport theory, Wasserstein geometry and entropic regularization, leveraging measure theory, convex optimization, PDEs for flow and diffusion models and connections to Reinforcement Learning.

Theoretical Computer Vision research guided by Prof. Shashank Vatedka, Dept. of EE:

- Developing weakly supervised methods for building localization and segmentation from satellite imagery using sparse point annotations, leveraging Marked Point Processes, Gibbs Energy Models, and Bayesian inference via RJMCMC for structured prediction and optimization.

Deepfake Detection using Physiological Signal Processing guided by Prof. Sumohana S. Channappayya, Dept. of EE:

- Selected for the IITH BUILD Cohort 2025 for proposing a novel deepfake detection framework integrating physiological signal cues, GAN-based representations, Vision Transformers, and multimodal feature fusion for robust and interpretable detection.

RELEVANT COURSES

- **Mathematics**: Probability and Random Variables, Statistics, Random Processes, Concentration Inequalities, Convex Optimization, Information Theory, Calculus-I, Calculus-II, Discrete Maths, Differential Equations, Transform Techniques, Complex Variables, Algorithms, Data Structures, Linear Algebra, Numerical Analysis, Group Theory, Metric Spaces, Theory of Computation, Systems Biology.
- **Programming and CS:** Computer Architecture, OS-I, OS-II, DBMS-I, DBMS-II, Compilers-I, Intro to Programming,

Software Development Fundamentals, Digital Circuits.

SKILLS

- C++, C, Python, Java, Linux, Shell Scripting, SQL, MATLAB, RISC-V, PyTorch, TensorFlow, NumPy, Pandas, Scikit-learn

KEY PROJECTS

- **Sentiment Analysis from IMDB Movie Reviews:** Developed RNN/LSTM-based sentiment classification models with Bahdanau and Luong attention mechanisms, leveraging pre-trained GloVe embeddings, attention visualization, and comprehensive benchmarking for improved interpretability and performance. GitHub
- **Multi-Modal Candidate Shortlisting System:** Developed a multimodal recruitment intelligence framework integrating BERT-based text representations with tabular neural networks for candidate classification, achieving 91% F1-score and 0.75 ROC-AUC. GitHub
- **Wireless Signal Classification and Interference Detection:** Developed MobileNetV3-based deep learning models for spectrogram classification, leveraging transfer learning, SpecAugment, MixUp, AdamW optimization, and cosine annealing schedules to achieve 90% accuracy; deployed models using ONNX for efficient inference. GitHub
- **Learning Biological Dynamics with Neural Networks**: Did as Systems Biology Course project. Developed a Scientific Machine Learning framework integrating nonlinear ODE-based glucose-insulin models with SBINNs for parameter estimation, structural identifiability and uncertainty quantification. GitHub

POSITIONS OF RESPONSIBILITY

Core member at Epoch - the AI, ML and Data Science club of IITH: Worked on candidate shortlisting system and sentiment analysis in learning phase and did quantum biologically inspired music generation engine as core project.

TEACHING EXPERIENCE AND VOLUNTEERING

- Teaching Assistant for **Introduction to Programming(ID1063)**, **Probability and Random Variables(AI1110), Random Processes(EE2080), ACM Summer School on Symmetric-Key Cryptography**.
- Volunteer for organizing **4th International Conference in Recent Advances in Applied Mathematics**

CERTIFICATIONS

Completed DeepLearning.AI specializations in Machine Learning, Deep Learning, NLP, GANs, and AI for Medicine, involving hands-on projects on neural networks, transformers, generative models, computer vision, and medical AI applications.